Wisdom & Chance TCG

Mobile Sync Guide & Feature Roadmap

API Version 2.3.0 | March 2026

For the Mobile Development Team

Base URL: https://wisdom-and-chance.replit.app
WebSocket: wss://wisdom-and-chance.replit.app/ws

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1. Current Feature Inventory

1.1 Core Game Systems

Feature	Status	Platform	Notes
200 Unit Cards (5 elements)	Live	Web + API	Power 1-10, traits, buffs/debuffs
10 Commanders	Live	Web + API	4 ability categories
Deck Builder	Live	Web + API	40 cards, power distribution rules
Practice vs AI	Live	Web + API	Easy/Medium/Hard difficulty
Card Database	Live	Web + API	Browse all cards with filters
Card Art (AI)	Live	Admin	Gemini 2.5 Flash image generation
Rarity System	Live	Web + API	Common/Rare/Epic/Legendary from power

1.2 Multiplayer & Social

Feature	Status	Platform	Notes
Real-time PvP	Live	Web + WS	Server-authoritative engine
Game Rooms	Live	Web + API + WS	Public/private, spectator support
Matchmaking	Live	Web + WS	ELO-based queue
In-game Chat	Live	Web + WS	Room + game chat
Spectator Mode	Live	Web + WS	Watch live games
Friend System	Live	Web + API	Requests, online status
Emotes	Live	Web + WS	In-game reactions

1.3 Progression & Engagement

Feature	Status	Platform	Notes
Achievements	Live	Web + API	5 categories, XP + gold rewards
Daily Challenges	Live	Web + API	Rotating daily goals
Leaderboard	Live	Web + API	ELO tiers: Bronze to Master
Player Stats	Live	Web + API	Level, XP, streaks, favorites
Economy System	Built	Web + API	Feature-flagged, ready to enable
Collection System	Built	Web + API	Card ownership, starter grants

1.4 Infrastructure

Feature	Status	Notes
Unified Auth	Live	Session cookies (web) + JWT Bearer (mobile)
Mobile Auth	Live	POST /api/mobile/auth/login with JWT (7-day)
Feature Flags	Live	Server-controlled, 15+ flags
API Docs	Live	GET /api/docs — full JSON reference
Admin Sync	Live	GET /api/admin/sync (code-protected)
PWA Support	Live	Manifest, service worker, icons
Database Backups	Live	Admin export + pg_dump
Health Check	Live	GET /api/health

2. Gap Analysis vs Major CCGs

Comparison against Hearthstone, MTG Arena, and Master Duel. Features are categorized by implementation priority.

2.1 Tier 1 — Critical (Core Progression Loop)

Feature	Hearthstone	MTG Arena	Master Duel	W&C Status
Card Rarity	Yes (6 tiers)	Yes (4 tiers)	Yes (4 tiers)	DONE (4 tiers)
Card Collection	Yes	Yes	Yes	DONE (flagged)
In-game Currency	Gold/Dust	Gold/Gems	Gems/CP	DONE (flagged)
Pack Opening	Animated	Animated	Animated	Basic (next)
Crafting/Disenchant	Dust system	Wildcards	CP crafting	DONE (flagged)
Shop / Store	Yes	Yes	Yes	Not built (next)
Starter Decks	Free decks	Color decks	Starter + Solo	Starter done

2.2 Tier 2 — High (Daily Engagement)

Feature	Hearthstone	MTG Arena	Master Duel	W&C Status
Seasons	Monthly	Monthly	Monthly	Planned
Battle Pass	Tavern Pass	Mastery Pass	Duel Pass	Planned
Daily Login	Quest-based	Daily rewards	Daily login	Not built
Match History	Recent games	Yes	Duel Log	Not built
Tutorials	Interactive	Color challenge	Solo Mode	Basic

2.3 Tier 3 — Medium (Collectibility & Monetization)

Feature	Hearthstone	MTG Arena	Master Duel	W&C Status
Card Variants	Golden/Diamond	Styles/Alt art	Royal/Prismatic	Not built
Cosmetics	Hero skins	Pets/Sleeves	Mates/Fields	Not built
Premium Currency	Runestones	Gems	Gems	Gems reserved
Tournaments	Battlegrounds	Events	Duelist Cup	Not built
Draft/Arena	Arena mode	Draft/Sealed	N/A	Not built

2.4 Tier 4 — Nice to Have

Feature	Status	Notes
Sound/Music	Not built	Audio engine + asset pipeline
Card Animations	Not built	Play/attack/death VFX
Guilds/Alliances	Not built	Social feature
Limited-time Modes	Not built	Seasonal game variants
Replays	Not built	Match recording/playback

3. Server-Sync Architecture Guide

3.1 Thin Client Principles

The mobile app should be a thin client that relies on the server as the single source of truth for all game state and data.

- Never trust client-side calculations for currency, collection, or game state
- All mutations go through server API — server validates and returns canonical state
- Card data, commanders, and game constants come from the server (cache locally)
- Feature flags control what UI to show — always respect GET /api/config
- Game logic is server-authoritative — clients send actions, server resolves them

3.2 Feature Flags

Every major feature is gated by a boolean flag returned from GET /api/config. The mobile app MUST check these before rendering UI sections.

```
GET /api/config !’ response.features
{
  "economy_enabled": false,    // Shop, currencies, collection
  "practice_mode": true,       // Practice vs AI
  "multiplayer": true,         // PvP, rooms, matchmaking
  "friends_system": true,      // Friend requests, status
  "daily_challenges": true,    // Daily goals
  "achievements": true,        // Achievement tracking
  "leaderboard": true,         // ELO rankings
  "spectator_mode": true,      // Watch live games
  "ranked_seasons": false,     // Seasonal ranking
  "battle_pass": false,        // Season pass rewards
  "shop_enabled": false,       // In-game shop
  "emotes": false              // In-game reactions
}
```

3.3 Versioned API

The server returns apiVersion in GET /api/config. The mobile app should:

- Store its own minimum supported API version
- Compare against server's apiVersion on launch
- If server version < mobile minimum, show 'server update pending' notice
- If mobile version < server's minClientVersion, force app update prompt

3.4 Authentication Flow

```
1. Mobile calls POST /api/mobile/auth/login { email, provider }
2. Server returns { token: "JWT...", user: {...} }
3. Store JWT securely (Keychain/Keystore)
4. Include "Authorization: Bearer <token>" on all requests
5. Token expires in 7 days
6. Before expiry: POST /api/mobile/auth/refresh !’ new token
7. On 401: redirect to login screen
```

3.5 Data Sync Strategy

Recommended caching and sync approach for mobile:

Data Type	Cache Duration	Sync Trigger	Strategy
Cards/Commanders	24 hours	App launch	Cache aggressively, rarely changes
Config/Flags	5 minutes	App launch + resume	Short cache, affects UI
Collection	On mutation	Pack open, craft, disenchant	Invalidate on write
Currencies	On mutation	Any economy action	Invalidate on write
Decks	On mutation	Create/edit/delete	Invalidate on write
Friends	2 minutes	Friend list view	Moderate cache
Leaderboard	5 minutes	Leaderboard view	Moderate cache
Game State	Never cache	WebSocket real-time	Always live via WS

4. API Contracts & Versioning

4.1 Base Configuration

```
Base URL: https://wisdom-and-chance.replit.app
API Prefix: /api
Auth Header: Authorization: Bearer <jwt_token>
Content-Type: application/json
Current Version: 2.3.0
```

4.2 Authentication Endpoints

Method	Path	Auth	Description
POST	/api/mobile/auth/login	No	Get JWT token (7-day expiry)
POST	/api/mobile/auth/refresh	Yes	Refresh JWT token
GET	/api/mobile/auth/me	Yes	Get current user info
GET	/api/auth/user	Yes	Get auth user (both web/mobile)

4.3 Public Endpoints (No Auth)

Method	Path	Description
GET	/api/health	Health check + DB status
GET	/api/config	Feature flags, version, maintenance
GET	/api/cards	All 200 cards with rarity
GET	/api/cards/:id	Single card by ID
GET	/api/cards/element/:element	Cards by element
GET	/api/commanders	All 10 commanders
GET	/api/commanders/:id	Single commander
GET	/api/achievements	All achievements
GET	/api/daily-challenges	Today's challenges
GET	/api/leaderboard	Top 100 by ELO
GET	/api/rooms	Public waiting rooms
GET	/api/docs	Full API reference (JSON)

4.4 Protected Endpoints (Auth Required)

Method	Path	Description
PATCH	/api/user/profile	Update profile info
GET	/api/user-decks	List user's saved decks
POST	/api/user-decks	Create deck (40 cards, validated)
PATCH	/api/user-decks/:id	Update deck
DELETE	/api/user-decks/:id	Delete deck
GET	/api/player-stats	Player stats (level, XP, streaks)
GET	/api/player-achievements	User's achievement progress
GET	/api/player-challenges	User's challenge progress
POST	/api/player-challenges/:id/claim	Claim challenge reward

4.5 Multiplayer & Room Endpoints

Method	Path	Description
POST	/api/rooms	Create game room
POST	/api/rooms/:id/join	Join a room
POST	/api/rooms/:id/leave	Leave a room
POST	/api/rooms/:id/ready	Toggle ready status
POST	/api/rooms/:id/start	Start game (host only)
POST	/api/rooms/:id/spectate	Join as spectator
DELETE	/api/rooms/:id/spectate	Leave spectating
GET	/api/rooms/:id/messages	Get room chat history
POST	/api/rooms/:id/messages	Send room chat message
GET	/api/live-matches	Active matches for spectating

4.6 Friend System Endpoints

Method	Path	Description
GET	/api/friends	List accepted friends
GET	/api/friend-requests	List pending requests
POST	/api/friend-requests	Send friend request
POST	/api/friend-requests/:id/accept	Accept request
POST	/api/friend-requests/:id/decline	Decline request

5. Economy System (Implemented)

The economy system is fully built and feature-flagged behind 'economy_enabled'. When enabled, it adds currencies, card ownership, pack opening, crafting, and disenchanting.

5.1 Rarity System

Rarity	Power Range	Pack Weight	Craft Cost (Dust)	Disenchant (Dust)
Common	1-3	60%	40	5
Rare	4-6	25%	100	20
Epic	7-8	10%	400	100
Legendary	9-10	5%	1,600	400

5.2 Currency Constants

Constant	Value	Notes
Starter Gold	500	Granted on first login
Pack Cost	100 gold	5 cards per pack
Match Win	30 gold	Auto-granted via WebSocket
Match Loss	10 gold	Auto-granted via WebSocket
Match Draw	15 gold	Auto-granted via WebSocket
Forfeit Win	15 gold	Opponent disconnected
Daily Challenge	25 gold	On claim (economy_enabled)
Achievement	50 gold	One-time per achievement

5.3 Economy Endpoints

Method	Path	Description
GET	/api/currencies	Get gold/gems/dust balances
GET	/api/collection	Get owned cards with quantities
POST	/api/packs/open	Buy & open pack (100 gold)
POST	/api/cards/craft	Craft card with dust
POST	/api/cards/disenchant	Disenchant for dust
POST	/api/collection/starter	Claim starter (idempotent)
POST	/api/achievements/:id/claim	Claim achievement gold

5.4 Starter Collection

On first login (web or mobile), players receive:

- 500 Gold
- 2 copies of every power 1-5 card (Common + Rare) across all 5 elements
- This is tracked by a 'starterClaimed' flag — calling /api/collection/starter is idempotent

5.5 Pack Opening Response

```
POST /api/packs/open
Response: {
  "cards": [
    { "cardId": "card-fire-3-2", "rarity": "Common", "isNew": true },
    { "cardId": "card-water-7-1", "rarity": "Epic", "isNew": false },
    ...
  ],
  "costGold": 100,
  "remainingGold": 400
}
```

5.6 Deck Ownership Validation

When economy is enabled, deck create/update validates that the player owns sufficient copies of each card. The server rejects decks containing cards the player doesn't own.

6. Upcoming Features & Roadmap

6.1 Next Up: Shop & Pack Opening Experience

Feature flag: shop_enabled, pack_opening_enabled

- Visual shop page with pack types and pricing
- Pack opening animation (card flip reveal with rarity glow)
- Multiple pack types: Basic (100g), Premium (250g), Element (150g)
- Purchase history and pack opening log
- API: GET /api/shop/packs, POST /api/shop/buy-pack

6.2 Ranked Seasons

Feature flag: ranked_seasons

- Monthly seasons with themes (e.g., 'Season 1: Flames of Dawn')
- ELO soft-reset at season end (50% toward 1000)
- Season-end rewards based on final tier
- Season countdown timer synced with server time
- API: GET /api/seasons/current, GET /api/seasons/:id/rewards

6.3 Battle Pass

Feature flag: battle_pass

- 30-tier reward track per season
- Free track: Gold, basic packs, common/rare cards
- Premium track: More gold, premium packs, exclusive cards
- XP from matches, challenges, achievements drives progression
- API: GET /api/battle-pass, POST /api/battle-pass/claim/:tier

6.4 Daily Login Rewards

7-day reward cycle with escalating rewards:

- Day 1: 10 Gold !' Day 7: Premium Pack + 50 Gold
- Streak counter and calendar UI
- API: POST /api/daily-login/claim

6.5 Card Variants & Cosmetics

Future monetization layer:

- Foil, animated, and alt-art card variants
- Same stats, different visuals
- Obtainable through packs, crafting, or shop
- Premium currency (Gems) for direct purchase

6.6 Implementation Priority Order

Phase	Features	Estimated Timeline	Dependencies
Phase 1	Shop + Pack Animation	Next sprint	Economy system (done)
Phase 2	Seasons + Battle Pass	Following sprint	Shop (phase 1)
Phase 3	Daily Login + Variants	After phase 2	Seasons (phase 2)
Phase 4	Tournaments + Draft	Future	Stable economy

7. WebSocket Contract

7.1 Connection

```
// Mobile WebSocket connection
const ws = new WebSocket(
  "wss://wisdom-and-chance.replit.app/ws?token=<jwt_token>"
);

// Web uses session cookies automatically
// Mobile MUST pass JWT via query parameter
```

7.2 Client !’ Server Events (from /api/docs)

Event	Payload	Description
join_room	{ roomId }	Subscribe to room updates
leave_room	{ roomId }	Unsubscribe from room
join_game	{ gameId }	Subscribe to game state updates
leave_game	{ gameId }	Unsubscribe from game updates
room_message	{ roomId, message }	Send chat message in room
game_message	{ gameId, message }	Send chat message in game
game_action	{ gameId, action }	Broadcast game state changes

Additional events supported by the server (not in /api/docs but functional):

- player_ready — { roomId, ready, deckId? } — toggle ready + select deck
- game_start — { roomId } — request game start (host)
- room_update — { roomId } — request room state refresh
- ping — {} — heartbeat keepalive
- auth — { token } — authenticate (alternative to query param)

7.3 Server !’ Client Events (from /api/docs)

Event	Payload	Description
auth_success	{ userId, displayName }	Auth confirmed, ready to send
auth_error	{ message }	Auth failed, will disconnect
room_update	Room state object	Room state changed
player_joined	{ roomId, userId }	Player joined room
player_left	{ roomId, userId }	Player left room
player_ready_update	{ roomId, userId, ready, deckId }	Ready status changed
game_start	{ roomId, gameId }	Game started — navigate to game
game_action	Game action object	Game state update from other player
chat_message	{ roomId, senderId, message, createdAt }	Chat message in room
game_message	{ userId, displayName, message, timestamp }	Chat message in game
friend_message	{ id, senderId, receiverId, message }	Friend direct message
friend_request	{ requestId, senderId }	Incoming friend request
friend_request_accepted	{ requestId, friendId }	Request accepted
presence_update	{ userId, status }	Friend online/offline
spectator_joined	{ roomId, userId }	Spectator joined
spectator_left	{ roomId, userId }	Spectator left

Additional server events (not in /api/docs but emitted by the engine):

- game_state — full sanitized game state (per-player view)
- game_over — { winner, reason, goldReward? } — game ended + rewards
- error — { message } — server error notification

7.4 Game Action Types

```
game_action payload:
{ gameId: "uuid", action: {
  type: "draw" | "deploy" | "end_turn" | "use_ability" | "forfeit",
  ...action-specific fields
}}

draw:        { type: "draw" }
deploy:      { type: "deploy", cardId: "card-fire-3-2", slot: 0-3 }
use_ability: { type: "use_ability" }
end_turn:    { type: "end_turn" }
forfeit:     { type: "forfeit" }
```

7.5 Reconnection Handling

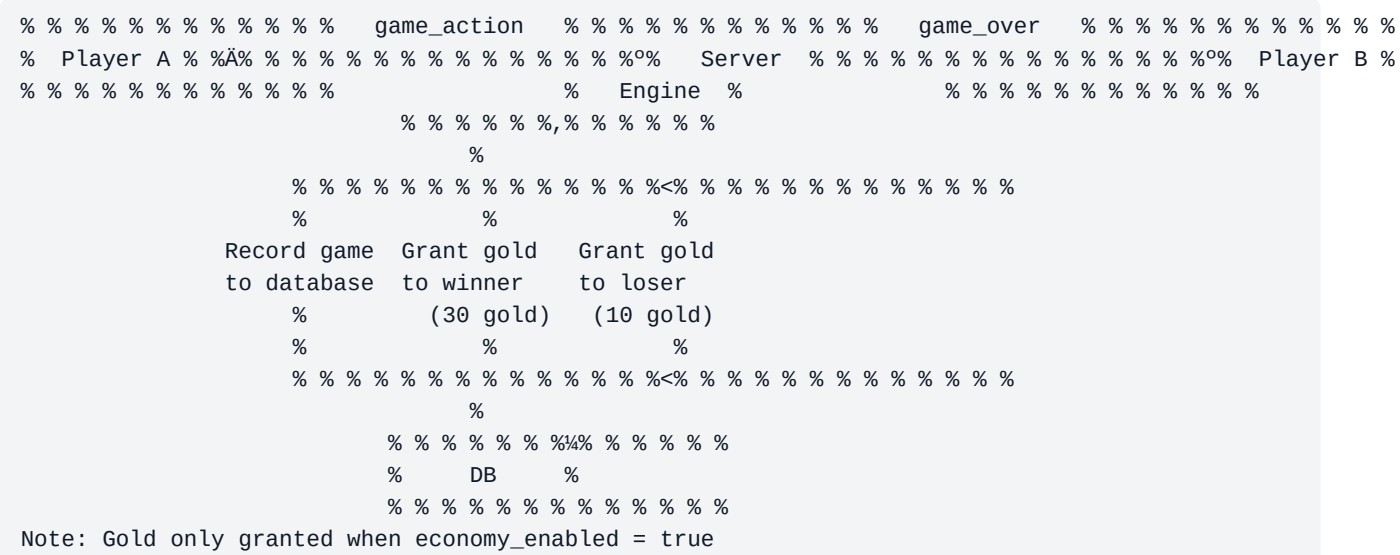
The server allows 60-second reconnection window for disconnected players:

- If a player disconnects during a game, their opponent gets a 60s countdown
- Reconnecting within 60s resumes the game — server sends full game_state
- After 60s, the disconnected player auto-forfeits
- Mobile should implement exponential backoff reconnect (1s, 2s, 4s, 8s, max 30s)

8. Data Flow Diagrams

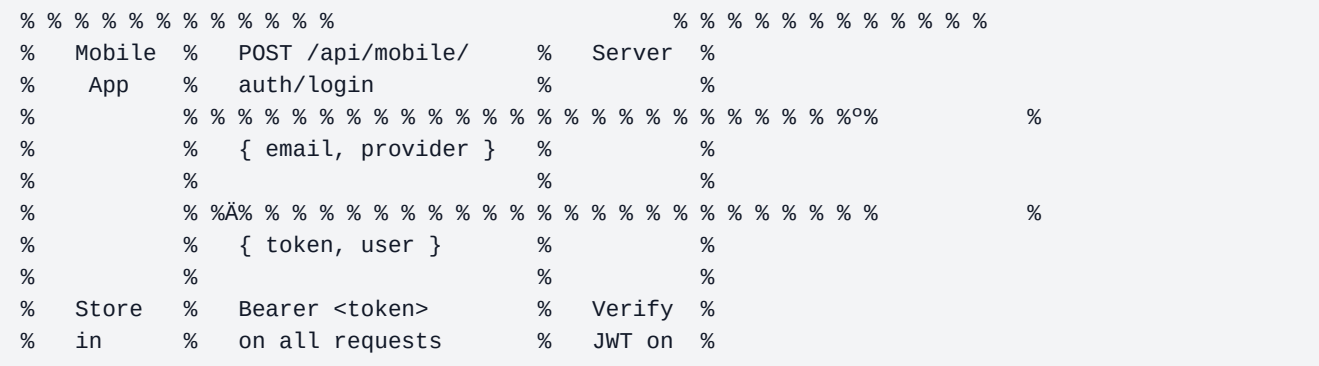
8.1 Economy Transaction Flow

8.2 Match Reward Flow



Note: Gold only granted when economy_enabled = true

8.3 Authentication Flow



% Keychain% %°% every %

```
%           %                       %   request %
```

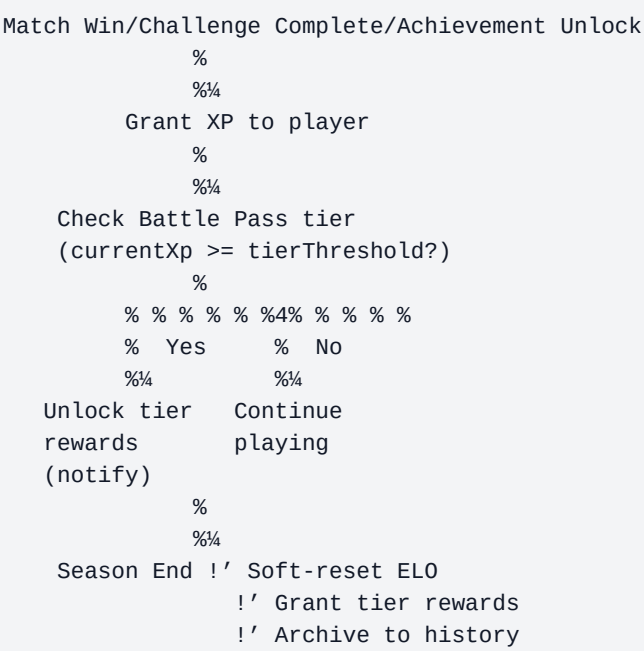
% % % % % % % % % % % %

% % % % % % % % % % % %

Token refresh: POST /api/mobile/auth/refresh

Returns new 7-day token. Call before expiry.

8.4 Season Progression (Planned)



9. Mobile-Specific Recommendations

9.1 Offline Caching Strategy

- Cache card data and commanders in local storage (SQLite/AsyncStorage)
- Cache user's decks for offline viewing (not editing)
- Cache leaderboard and friend list with timestamps
- Show cached data immediately, refresh in background
- Never cache game state — always require live connection for play
- Cache feature flags for 5 minutes max

9.2 Push Notification Hooks

The server can be extended with push notification triggers for:

- Friend request received
- Challenge/match invitation
- Daily challenge refresh (midnight UTC)
- Season ending soon (24h, 1h warnings)
- Battle pass tier unlocked
- Achievement completed

Mobile should register device token via future endpoint: POST /api/push/register

9.3 Reconnection Handling

- Implement WebSocket heartbeat (ping every 30s)
- Exponential backoff on disconnect: 1s !’ 2s !’ 4s !’ 8s !’ max 30s
- On reconnect during active game: server auto-sends full game_state
- 60-second window before auto-forfeit — show reconnection countdown to user
- On reconnect outside game: re-fetch /api/config, re-establish presence

9.4 Image Handling

- Card images available at: GET /api/cards/:id/image
- Commander images at: GET /api/commanders/:id/image
- Cache images aggressively — they rarely change
- Use placeholder/skeleton images while loading
- Consider progressive image loading for card database

9.5 Error Handling

Standard error response format:

```
{
  "message": "Human-readable error description",
  "error": "Error type identifier"  // optional
}

Common status codes:
  400 – Validation error (check message for details)
  401 – Token expired or invalid (redirect to login)
  403 – Not authorized for this resource
  404 – Resource not found
  429 – Rate limited (back off and retry)
  500 – Server error (show generic error, retry)
  503 – Maintenance mode (check /api/config)
```

9.6 Performance Tips

- Batch API calls on app launch: /api/config + /api/cards + /api/auth/user
- Use ETag/If-None-Match headers for card data caching
- Paginate leaderboard requests (server returns top 100)
- Debounce deck save calls (500ms delay after last edit)
- Pre-fetch next game state on opponent's turn

10. Appendix: Quick Reference

10.1 Card ID Format

```
card-{element}-{power}-{variant}
Examples: card-fire-3-2, card-water-7-1, card-earth-10-0
```

10.2 Element List

```
Fire | Water | Earth | Air | Nature
(5 elements, 40 cards each = 200 total)
```

10.3 Deck Validation Rules

Rule	Value
Total cards	Exactly 40
Cards per power rank	4 (ranks 1-10)
Max copies per card	3
Commander required	Yes (1 per deck)
Ownership check	When economy_enabled=true

10.4 ELO Rating Tiers

Tier	Rating Range
Bronze	< 800
Silver	800 – 999
Gold	1000 – 1199
Platinum	1200 – 1399
Diamond	1400 – 1599
Master	1600+

10.5 Game Constants

```
Starting HP: 40
Cards per hand: 5 (initial draw)
Deployment slots: 4 per side
Turn phases: Draw !’ Deploy !’ Combat !’ Calculate !’ End
Max power per card: 10
Buff/Debuff colors: Red, Blue, Green, Yellow, Purple
```

10.6 Key Server Files (For Reference)

File	Purpose
server/apiDocs.ts	Full API documentation (JSON)
server/routes.ts	All REST endpoints
server/websocket.ts	WebSocket event handling
server/gameEngine.ts	Server-authoritative game engine
server/economyService.ts	Economy helpers (grantGold, etc.)
server/multiplayerRoutes.ts	Multiplayer & social endpoints
server/mobileAuth.ts	Mobile JWT auth
shared/schema.ts	DB schema + types
shared/models/economy.ts	Economy tables + constants

10.7 Admin Endpoints (For Debugging)

Note: Admin endpoints require admin authentication. Contact the project owner for access codes.

Method	Path	Description
GET	/api/admin/sync	Full data dump (code-protected)
GET	/api/admin/feature-flags	View all feature flags
PATCH	/api/admin/feature-flags/:key	Toggle a feature flag
GET	/api/admin/database-export	Export all DB tables as JSON